

What's An Operating System?

Imagine you have a brand new computer. Imagine that nobody had put a disk of any kind into it, ever. That would mean that there was no software installed on the system. If you switched the computer on; It would beep a few times and then tell you that it couldn't start an operating system. The most important software to a computer is one thing: — the Operating System.

Without an Operating system, you couldn't surf the web, you couldn't play music, you couldn't write letters. You can't do anything.

Some of you will have heard of famous operating systems already but may not fully appreciate it. For example, Microsoft make a well known operating system called Windows, Apple make two that you may know: MacOS (on most Macintosh computers) and iOS (on iPhones and iPads). An operating system is the software that sits between you, the user, and the hardware inside the computer. If you click the mouse on an icon on your screen, the operating system interprets that you want to load the program that you are clicking on. For all of this to happen, The Operating system (some times referred to as the OS or O/S) must know how to use a screen (to show you what's going on), to use a mouse (so you can move it around and click with it), to use your hard disk drive (to load up the data from it). It must also need to know pretty much everything else about the hardware installed inside your computer, ie: RAM (Computer Memory), Hard disks/USB/CD drives, keyboards, joysticks, sound controllers, graphics controllers, printers, scanners, etc.

So when you start typing a letter, for example, you have already loaded up a word processing piece of software. This software is called application software and is running 'on top' of the Operating System, but nonetheless, all of the time whilst the word processor application is running, it talks constantly to the O/S for vital information.

Okay, we've established that an O/S is necessary, but what else does an O/S do: Probably the most basic and yet essential tasks of an operating system is the job of managing our files and data. A basic O/S should be able to do the following with files and folders:

- Create them
- Move them to other directories (otherwise known as folders)
- Rename them
- Copy them
- Delete/remove them
- Send and receive files to/from other devices such as Printers/Scanners and your Internet connection.
- ...and a bit more.

Now you have the idea of what an Operating system is, let's find out about a specific type of operating system called UNIX, the grandfather of Linux...